

# 08 Data Versioning and Lineage

**Course Name:** Data Management for Machine Learning

**Assignment Title:** End-to-End Data Management Pipeline for a Recommendation System

**Assignment:** Group 51 - Data management for Machine Learning Group 51

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## 1. Objective

This report documents the data versioning and lineage stage of the recommendation pipeline. It captures repository structure, discovered dataset versions, lineage metadata, and workflow documentation for reproducibility and auditability.

## 2. Objective Coverage

- Detect DVC or Git-based versioning assets where present.
- Register raw and downstream datasets with lineage metadata such as source, ingestion date, and transformations.
- Generate repository structure views showing dataset locations and version-related artifacts.
- Produce documentation of the versioning workflow for submission.

## 3. Supporting Project and Log Artifacts

The following discovered files support this section, including validation outputs, lineage registry artifacts, and workflow documentation. Long paths are wrapped using real line breaks only at safe separators to avoid cut-off text in the PDF.

| Log / Report File                                   | Relative Path                                                          | Last Modified       | Size     |
|-----------------------------------------------------|------------------------------------------------------------------------|---------------------|----------|
| run_validation.txt                                  | src\02-validation\run_validation.txt                                   | 2026-04-28 20:14:12 | 46.94 KB |
| data_quality_report_20260429T111342Z.json           | reports\validation\data_quality_report_20260429T111342Z.json           | 2026-04-29 16:47:43 | 6.66 KB  |
| data_quality_report_20260429T111342Z.pdf            | reports\validation\data_quality_report_20260429T111342Z.pdf            | 2026-04-29 16:47:44 | 8.49 KB  |
| fix_log_20260429T111342Z.csv                        | reports\validation\fix_log_20260429T111342Z.csv                        | 2026-04-29 16:46:38 | 3.43 KB  |
| validation_summary_revalidated_20260429T111342Z.csv | reports\validation\validation_summary_revalidated_20260429T111342Z.csv | 2026-04-29 16:47:43 | 0.41 KB  |

| Log / Report File                                  | Relative Path                                                         | Last Modified       | Size     |
|----------------------------------------------------|-----------------------------------------------------------------------|---------------------|----------|
| validation_issues_revalidated_20260429T111342Z.csv | reports\validation\validation_issues_revalidated_20260429T111342Z.csv | 2026-04-29 16:47:43 | 0.51 KB  |
| validation_log_20260429T111342Z.jsonl              | logs\validation_log_20260429T111342Z.jsonl                            | 2026-04-29 16:47:44 | 0.66 KB  |
| dataset_lineage_registry.csv                       | data_versioning\registry\dataset_lineage_registry.csv                 | 2026-04-29 22:33:24 | 7.91 KB  |
| dataset_lineage_registry.json                      | data_versioning\registry\dataset_lineage_registry.json                | 2026-04-29 22:33:24 | 19.28 KB |
| versioning_workflow.md                             | data_versioning\docs\versioning_workflow.md                           | 2026-04-29 22:33:24 | 1.7 KB   |
| versioning_config.json                             | data_versioning\registry\versioning_config.json                       | 2026-04-29 22:33:24 | 0.96 KB  |
| data_versioning_log_20260429T170315Z.jsonl         | logs\data_versioning_log_20260429T170315Z.jsonl                       | 2026-04-29 22:33:24 | 0.54 KB  |

## 4. Versioning and Lineage Assets in Project

| Versioning / Lineage Asset           | Relative Path                                          | Last Modified       | Size     |
|--------------------------------------|--------------------------------------------------------|---------------------|----------|
| .dvc                                 | .dvc                                                   | 2026-04-28 15:16:13 | 0.0 KB   |
| config                               | .dvc\config                                            | 2026-04-28 15:14:51 | 0.08 KB  |
| .dvcignore                           | .dvcignore                                             | 2026-04-28 15:12:07 | 0.14 KB  |
| prepared.dvc                         | data\prepared.dvc                                      | 2026-04-28 15:17:43 | 0.11 KB  |
| raw.dvc                              | data\raw.dvc                                           | 2026-04-28 15:16:13 | 0.1 KB   |
| models.dvc                           | models.dvc                                             | 2026-04-28 15:16:22 | 0.1 KB   |
| 08 Data Versioning and Lineage.ipynb | src\07-submission\08 Data Versioning and Lineage.ipynb | 2026-04-29 22:33:16 | 53.88 KB |

## 5. Git / Versioning Summary

| head_ref        | commit_hash                              | dvc_pointer_files | datasets_registered |
|-----------------|------------------------------------------|-------------------|---------------------|
| refs/heads/main | 5ce2fe6dbb26109f4dfdc645cb4392aff8253e64 | 3                 | 32                  |

## 6. DVC Pointer Summary

| DVC Pointer       | Tracked Output Path | MD5 | Size   |
|-------------------|---------------------|-----|--------|
| data\prepared.dvc | prepared            |     | 329    |
| data\raw.dvc      | raw                 |     | 446229 |
| models.dvc        | models              |     | 0      |

7. Dataset Lineage Summary

| metric                       | value |
|------------------------------|-------|
| datasets_discovered          | 32    |
| raw_datasets                 | 6     |
| bronze_datasets              | 7     |
| prepared_processed_datasets  | 7     |
| transformed_feature_datasets | 4     |
| remediated_datasets          | 8     |
| dvc_tracked_datasets         | 0     |
| unique_source_systems        | 3     |

## 8. Dataset Lineage Registry Preview

The registry below documents dataset name, stage, source, inferred ingestion date, applied transformations, and version label.

| dataset_name                                  | stage    | source_system | relative_path                                                            | ingestion_date_inferred | applied_transformations | version_label |
|-----------------------------------------------|----------|---------------|--------------------------------------------------------------------------|-------------------------|-------------------------|---------------|
| categories.parquet                            | bronze   | dummyjson     | data\bronze\dummyjson\categories.parquet                                 |                         | format_standardization  | unversioned   |
| ingestion_summary.csv                         | bronze   | dummyjson     | data\bronze\dummyjson\ingestion_summary.csv                              |                         | format_standardization  | unversioned   |
| products.parquet                              | bronze   | dummyjson     | data\bronze\dummyjson\products.parquet                                   |                         | format_standardization  | unversioned   |
| category_tree.parquet                         | bronze   | retailrocket  | data\bronze\retailrocket\category_tree.parquet                           |                         | format_standardization  | unversioned   |
| events.parquet                                | bronze   | retailrocket  | data\bronze\retailrocket\events.parquet                                  |                         | format_standardization  | unversioned   |
| ingestion_summary.csv                         | bronze   | retailrocket  | data\bronze\retailrocket\ingestion_summary.csv                           |                         | format_standardization  | unversioned   |
| item_properties.parquet                       | bronze   | retailrocket  | data\bronze\retailrocket\item_properties.parquet                         |                         | format_standardization  | unversioned   |
| retailrocket_item_snapshot.csv                | prepared | retailrocket  | data\prepared\retailrocket_item_snapshot.csv                             |                         | cleaning_preprocessing  | unversioned   |
| categories_prepared.csv                       | prepared | unknown       | data\prepared\categories_prepared.csv                                    |                         | cleaning_preprocessing  | unversioned   |
| interactions_prepared.csv                     | prepared | unknown       | data\prepared\interactions_prepared.csv                                  |                         | cleaning_preprocessing  | unversioned   |
| prepared_interactions_20260429_215146.parquet | prepared | unknown       | data\prepared\prepared_interactions_20260429_215146.parquet              |                         | cleaning_preprocessing  | unversioned   |
| prepared_interactions_20260429_215229.parquet | prepared | unknown       | data\prepared\prepared_interactions_20260429_215229.parquet              |                         | cleaning_preprocessing  | unversioned   |
| prepared_interactions_20260429_215532.parquet | prepared | unknown       | data\prepared\prepared_interactions_20260429_215532.parquet              |                         | cleaning_preprocessing  | unversioned   |
| products_prepared.csv                         | prepared | unknown       | data\prepared\products_prepared.csv                                      |                         | cleaning_preprocessing  | unversioned   |
| categories_raw.json                           | raw      | dummyjson     | data\raw\dummyjson\load_date=2026-04-29\load_hour=11\categories_raw.json | 2026-04-29              | initial_ingestion       | unversioned   |
| products_raw.json                             | raw      | dummyjson     | data\raw\dummyjson\load_date=2026-04-29\load_hour=11\products_raw.json   | 2026-04-29              | initial_ingestion       | unversioned   |

| dataset_name              | stage      | source_system | relative_path                                                                     | ingestion_date_inferred | applied_transformations | version_label |
|---------------------------|------------|---------------|-----------------------------------------------------------------------------------|-------------------------|-------------------------|---------------|
| category_tree.csv         | raw        | retailrocket  | data\raw\retailrocket\load_date=2026-04-29\load_hour=11\category_tree.csv         | 2026-04-29              | initial_ingestion       | unversioned   |
| events.csv                | raw        | retailrocket  | data\raw\retailrocket\load_date=2026-04-29\load_hour=11\events.csv                | 2026-04-29              | initial_ingestion       | unversioned   |
| item_properties_part1.csv | raw        | retailrocket  | data\raw\retailrocket\load_date=2026-04-29\load_hour=11\item_properties_part1.csv | 2026-04-29              | initial_ingestion       | unversioned   |
| item_properties_part2.csv | raw        | retailrocket  | data\raw\retailrocket\load_date=2026-04-29\load_hour=11\item_properties_part2.csv | 2026-04-29              | initial_ingestion       | unversioned   |
| categories.parquet        | remediated | unknown       | data\remediated\bronze\categories.parquet                                         |                         | automated_remediation   | unversioned   |
| categories_raw.json       | remediated | unknown       | data\remediated\raw\categories_raw.json                                           |                         | automated_remediation   | unversioned   |
| category_tree.csv         | remediated | unknown       | data\remediated\raw\category_tree.csv                                             |                         | automated_remediation   | unversioned   |
| events.csv                | remediated | unknown       | data\remediated\raw\events.csv                                                    |                         | automated_remediation   | unversioned   |
| item_properties_part1.csv | remediated | unknown       | data\remediated\raw\item_properties_part1.csv                                     |                         | automated_remediation   | unversioned   |

## 9. Versioning Workflow Summary

- Raw datasets are discovered from partitioned source folders under data/raw.
- Bronze, prepared, transformed, features, and remediated folders are registered as downstream lineage stages.
- load\_date and load\_hour are inferred from partitioned folder names when present.
- A SHA256 hash is recorded for each discovered dataset to support traceability.
- If a matching .dvc pointer exists, the dataset is labeled as DVC-tracked.
- If no DVC pointer exists, the notebook still builds a custom lineage registry for documentation and audit use.
- Validation outputs and fix logs are included as supporting evidence in reports/validation and logs.

## 10. Repository Structure Showing Dataset Versions

### Project Root Structure

```

recomart-pipeline/
├── .dvc/
├── cache/
├── files/
├── tmp/
├── btime
└── lock

```

```
■ ■ ■■■ rwlock
■ ■ ■■■ rwlock.lock
■ ■■■ .gitignore
■ ■■■ config
■■■ .git/
■ ■■■ hooks/
■ ■ ■■■ applypatch-msg.sample
■ ■ ■■■ commit-msg.sample
■ ■ ■■■ fsmonitor-watchman.sample
■ ■ ■■■ post-checkout
■ ■ ■■■ post-commit
■ ■ ■■■ post-merge
■ ■ ■■■ post-update.sample
■ ■ ■■■ pre-applypatch.sample
■ ■ ■■■ pre-commit.sample
■ ■ ■■■ pre-merge-commit.sample
■ ■ ■■■ pre-push
■ ■ ■■■ pre-push.sample
■ ■ ■■■ pre-rebase.sample
■ ■ ■■■ pre-receive.sample
■ ■ ■■■ prepare-commit-msg.sample
■ ■ ■■■ push-to-checkout.sample
■ ■ ■■■ sendemail-validate.sample
■ ■ ■■■ update.sample
■ ■■■ info/
■ ■ ■■■ exclude
■ ■■■ lfs/
■ ■ ■■■ cache/
■ ■ ■■■ objects/
■ ■ ■■■ tmp/
■ ■■■ logs/
■ ■ ■■■ refs/
■ ■ ■■■ HEAD
■ ■■■ objects/
■ ■ ■■■ 00/
■ ■ ■■■ 01/
■ ■ ■■■ 02/
■ ■ ■■■ 04/
■ ■ ■■■ 05/
■ ■ ■■■ 06/
■ ■ ■■■ 07/
■ ■ ■■■ 08/
■ ■ ■■■ 09/
■ ■ ■■■ 0b/
■ ■ ■■■ 0c/
■ ■ ■■■ 0d/
■ ■ ■■■ 0e/
■ ■ ■■■ 0f/
■ ■ ■■■ 11/
■ ■ ■■■ 12/
■ ■ ■■■ 14/
■ ■ ■■■ 16/
■ ■ ■■■ 17/
■ ■ ■■■ 18/
■ ■ ■■■ 19/
■ ■ ■■■ 1a/
■ ■ ■■■ 1b/
■ ■ ■■■ 1c/
■ ■ ■■■ 1d/
■ ■ ■■■ 1e/
■ ■ ■■■ 1f/
■ ■ ■■■ 20/
■ ■ ■■■ 21/
■ ■ ■■■ 23/
■ ■ ■■■ 24/
■ ■ ■■■ 25/
■ ■ ■■■ 26/
■ ■ ■■■ 27/
■ ■ ■■■ 28/
■ ■ ■■■ 29/
■ ■ ■■■ 2a/
■ ■ ■■■ 2b/
■ ■ ■■■ 2d/
■ ■ ■■■ 2f/
■ ■ ■■■ 31/
■ ■ ■■■ 32/
■ ■ ■■■ 33/
■ ■ ■■■ 34/
■ ■ ■■■ 35/
■ ■ ■■■ 36/
■ ■ ■■■ 37/
■ ■ ■■■ 38/
■ ■ ■■■ 39/
■ ■ ■■■ 3a/
■ ■ ■■■ 3b/
■ ■ ■■■ 3c/
■ ■ ■■■ 3d/
■ ■ ■■■ 3e/
■ ■ ■■■ 3f/
```

|   |   |     |     |
|---|---|-----|-----|
| ■ | ■ | ■■■ | 40/ |
| ■ | ■ | ■■■ | 41/ |
| ■ | ■ | ■■■ | 42/ |
| ■ | ■ | ■■■ | 44/ |
| ■ | ■ | ■■■ | 45/ |
| ■ | ■ | ■■■ | 46/ |
| ■ | ■ | ■■■ | 47/ |
| ■ | ■ | ■■■ | 48/ |
| ■ | ■ | ■■■ | 49/ |
| ■ | ■ | ■■■ | 4a/ |
| ■ | ■ | ■■■ | 4b/ |
| ■ | ■ | ■■■ | 4c/ |
| ■ | ■ | ■■■ | 4d/ |
| ■ | ■ | ■■■ | 4e/ |
| ■ | ■ | ■■■ | 4f/ |
| ■ | ■ | ■■■ | 50/ |
| ■ | ■ | ■■■ | 51/ |
| ■ | ■ | ■■■ | 52/ |
| ■ | ■ | ■■■ | 53/ |
| ■ | ■ | ■■■ | 54/ |
| ■ | ■ | ■■■ | 55/ |
| ■ | ■ | ■■■ | 56/ |
| ■ | ■ | ■■■ | 57/ |
| ■ | ■ | ■■■ | 58/ |
| ■ | ■ | ■■■ | 59/ |
| ■ | ■ | ■■■ | 5a/ |
| ■ | ■ | ■■■ | 5b/ |
| ■ | ■ | ■■■ | 5c/ |
| ■ | ■ | ■■■ | 5d/ |
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| ■ | ■ | ■■■ | 60/ |
| ■ | ■ | ■■■ | 61/ |
| ■ | ■ | ■■■ | 62/ |
| ■ | ■ | ■■■ | 63/ |
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| ■ | ■ | ■■■ | 66/ |
| ■ | ■ | ■■■ | 67/ |
| ■ | ■ | ■■■ | 68/ |
| ■ | ■ | ■■■ | 69/ |
| ■ | ■ | ■■■ | 6a/ |
| ■ | ■ | ■■■ | 6b/ |
| ■ | ■ | ■■■ | 6c/ |
| ■ | ■ | ■■■ | 6d/ |
| ■ | ■ | ■■■ | 6f/ |
| ■ | ■ | ■■■ | 70/ |
| ■ | ■ | ■■■ | 71/ |
| ■ | ■ | ■■■ | 72/ |
| ■ | ■ | ■■■ | 73/ |
| ■ | ■ | ■■■ | 74/ |
| ■ | ■ | ■■■ | 75/ |
| ■ | ■ | ■■■ | 77/ |
| ■ | ■ | ■■■ | 78/ |
| ■ | ■ | ■■■ | 79/ |
| ■ | ■ | ■■■ | 7a/ |
| ■ | ■ | ■■■ | 7b/ |
| ■ | ■ | ■■■ | 7c/ |
| ■ | ■ | ■■■ | 7d/ |
| ■ | ■ | ■■■ | 7e/ |
| ■ | ■ | ■■■ | 7f/ |
| ■ | ■ | ■■■ | 80/ |
| ■ | ■ | ■■■ | 81/ |
| ■ | ■ | ■■■ | 82/ |
| ■ | ■ | ■■■ | 83/ |
| ■ | ■ | ■■■ | 84/ |
| ■ | ■ | ■■■ | 85/ |
| ■ | ■ | ■■■ | 86/ |
| ■ | ■ | ■■■ | 87/ |
| ■ | ■ | ■■■ | 88/ |
| ■ | ■ | ■■■ | 89/ |
| ■ | ■ | ■■■ | 8b/ |
| ■ | ■ | ■■■ | 8c/ |
| ■ | ■ | ■■■ | 8d/ |
| ■ | ■ | ■■■ | 8e/ |
| ■ | ■ | ■■■ | 8f/ |
| ■ | ■ | ■■■ | 90/ |
| ■ | ■ | ■■■ | 91/ |
| ■ | ■ | ■■■ | 92/ |
| ■ | ■ | ■■■ | 93/ |
| ■ | ■ | ■■■ | 94/ |
| ■ | ■ | ■■■ | 95/ |
| ■ | ■ | ■■■ | 96/ |
| ■ | ■ | ■■■ | 97/ |
| ■ | ■ | ■■■ | 9a/ |
| ■ | ■ | ■■■ | 9b/ |
| ■ | ■ | ■■■ | 9c/ |
| ■ | ■ | ■■■ | 9d/ |
| ■ | ■ | ■■■ | 9f/ |
| ■ | ■ | ■■■ | a0/ |

```

■ ■ ■■■ a2/
■ ■ ■■■ a3/
■ ■ ■■■ a4/
■ ■ ■■■ a5/
■ ■ ■■■ a6/
■ ■ ■■■ a7/
■ ■ ■■■ a8/
■ ■ ■■■ a9/
■ ■ ■■■ aa/
■ ■ ■■■ ac/
■ ■ ■■■ ad/
■ ■ ■■■ ae/
■ ■ ■■■ af/
■ ■ ■■■ b0/
■ ■ ■■■ b1/
■ ■ ■■■ b2/
■ ■ ■■■ b3/
■ ■ ■■■ b4/
■ ■ ■■■ b5/
■ ■ ■■■ b6/
■ ■ ■■■ b8/
■ ■ ■■■ b9/
■ ■ ■■■ ba/
■ ■ ■■■ bb/
■ ■ ■■■ bc/
■ ■ ■■■ bd/
■ ■ ■■■ be/
■ ■ ■■■ bf/
■ ■ ■■■ c1/
■ ■ ■■■ c2/
■ ■ ■■■ c3/
■ ■ ■■■ c4/
■ ■ ■■■ c5/
■ ■ ■■■ c6/
■ ■ ■■■ c7/
■ ■ ■■■ c8/
... output truncated ...

```

## Raw Data Structure

```

raw/
■■■ dummyjson/
■ ■■■ load_date=2026-04-29/
■ ■■■ load_hour=11/
■ ■■■ categories_raw.json
■ ■■■ products_raw.json
■■■ retailrocket/
■■■ load_date=2026-04-29/
■ ■■■ load_hour=11/
■ ■■■ category_tree.csv
■ ■■■ events.csv
■ ■■■ item_properties_part1.csv
■ ■■■ item_properties_part2.csv

```

## Downstream Data Structure

```

bronze/
■■■ dummyjson/
■ ■■■ categories.parquet
■ ■■■ ingestion_summary.csv
■ ■■■ products.parquet
■■■ retailrocket/
■ ■■■ category_tree.parquet
■ ■■■ events.parquet
■ ■■■ ingestion_summary.csv
■ ■■■ item_properties.parquet

prepared/
■■■ categories_prepared.csv
■■■ interactions_prepared.csv
■■■ prepared_interactions_20260429_215146.parquet
■■■ prepared_interactions_20260429_215229.parquet
■■■ prepared_interactions_20260429_215532.parquet
■■■ products_prepared.csv
■■■ retailrocket_item_snapshot.csv

transformed/
■■■ item_cooccurrence.csv
■■■ item_features.csv
■■■ user_features.csv
■■■ user_item_features.csv

features/

remediated/
■■■ bronze/
■ ■■■ categories.parquet
■ ■■■ products.parquet
■■■ raw/

```



- categories\_raw.json
- category\_tree.csv
- events.csv
- item\_properties\_part1.csv
- item\_properties\_part2.csv
- products\_raw.json

## Versioning Artifact Structure

- data\_versioning/
  - docs/
    - ■■■ versioning\_workflow.md
  - registry/
    - dataset\_lineage\_registry.csv
    - dataset\_lineage\_registry.json
    - versioning\_config.json

## 11. Versioning Workflow Documentation

```
# Data Versioning and Lineage Workflow

## Objective
Version raw and transformed datasets, track lineage metadata, and document how datasets move
through the RecoMart pipeline.

## Recommended Workflow
1. Ingest source data into partitioned raw folders under data/raw using source and load
timestamp conventions.
2. Store standardized outputs in bronze and downstream prepared/transformed/features locations.
3. Version datasets with DVC where available, or maintain this custom registry with hashes and
metadata when DVC pointers are absent.
4. Record metadata including dataset name, path, source system, ingestion date, stage,
transformations, and file hash.
5. Preserve validation reports, fix logs, and revalidation summaries in reports/validation for
auditability.
6. Regenerate lineage_registry artifacts after each ingestion or transformation run.

## Project Observations
- Datasets discovered: 32
- Versioning assets discovered: 7
- DVC pointer files discovered: 3
- Git HEAD ref: refs/heads/main
- Git commit hash: 5ce2fe6dbb26109f4dfdc645cb4392aff8253e64

## Metadata Fields Captured
- dataset_name
- relative_path
- stage
- source_system
- file_format
- size_kb
- last_modified
- load_date / load_hour inferred from partitioned paths when present
- applied_transformations
- versioning_tool
- version_label
- dvc_pointer_file
- sha256

## Usage Notes
- Prefer DVC for large dataset versioning and reproducible pulls/checkouts.
- Use Git for code, metadata, and pointer files rather than large raw binaries.
- If Git LFS is present, restrict it to large binary artifacts that do not fit normal Git
workflows.
- Keep lineage registry generation as part of the pipeline so metadata stays current.
```

## 12. Versioning Configuration Preview

```
{
  "project": "RecoMart Data Versioning and Lineage",
  "created_ts": "2026-04-29T17:03:15.268474+00:00",
  "registry_csv": "data_versioning\\registry\\dataset_lineage_registry.csv",
  "registry_json": "data_versioning\\registry\\dataset_lineage_registry.json",
  "workflow_doc": "data_versioning\\docs\\versioning_workflow.md",
  "git": {
    "head_ref": "refs/heads/main",
    "commit_hash": "5ce2fe6dbb26109f4dfdc645cb4392aff8253e64"
  },
  "dvc_pointer_files": [
    {
      "dvc_file": "data\\prepared.dvc",
      "outs_path": "prepared",
      "md5": null,
      "size": "329"
    },
    {
      "dvc_file": "data\\raw.dvc",
      "outs_path": "raw",
      "md5": null,
      "size": "446229"
    },
    {
      "dvc_file": "models.dvc",
      "outs_path": "models",
      "md5": null,
      "size": "0"
    }
  ],
  "dataset_counts": {
    "remediated": 8,
    "bronze": 7,
    "prepared": 7,
    "raw": 6,
  }
}
```

```

    "transformed": 4
  }
}

```

### 13. Lineage Registry JSON Preview

```

[
  {
    "dataset_name": "categories.parquet",
    "relative_path": "data\\bronze\\dummyjson\\categories.parquet",
    "stage": "bronze",
    "source_system": "dummyjson",
    "file_format": "parquet",
    "size_kb": 6.1,
    "last_modified": "2026-04-29 16:42:35",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "format_standardization",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "bebe14b201ba39e3dfd2c5d86e73c5bae0e62511b4795f578ed7c879f5180a17"
  },
  {
    "dataset_name": "ingestion_summary.csv",
    "relative_path": "data\\bronze\\dummyjson\\ingestion_summary.csv",
    "stage": "bronze",
    "source_system": "dummyjson",
    "file_format": "csv",
    "size_kb": 0.23,
    "last_modified": "2026-04-29 16:42:35",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "format_standardization",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "976a5d3240029c01cf6dddf18b68ea52478bf51bfae22f2259e343c6c76bdf9af"
  },
  {
    "dataset_name": "products.parquet",
    "relative_path": "data\\bronze\\dummyjson\\products.parquet",
    "stage": "bronze",
    "source_system": "dummyjson",
    "file_format": "parquet",
    "size_kb": 73.89,
    "last_modified": "2026-04-29 16:42:35",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "format_standardization",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "3a15b4d890d89bf1eaa42b68f2c6ad3964b5867c9fb9eca63ea1e6f49cfa92c1"
  },
  {
    "dataset_name": "category_tree.parquet",
    "relative_path": "data\\bronze\\retailrocket\\category_tree.parquet",
    "stage": "bronze",
    "source_system": "retailrocket",
    "file_format": "parquet",
    "size_kb": 16.33,
    "last_modified": "2026-04-29 16:42:58",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "format_standardization",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "204f14e274fe7916a405278b3db42fa9d792bb2e6a18c742ee0f131793545ad7"
  },
  {
    "dataset_name": "events.parquet",
    "relative_path": "data\\bronze\\retailrocket\\events.parquet",
    "stage": "bronze",
    "source_system": "retailrocket",
    "file_format": "parquet",
    "size_kb": 56736.72,
    "last_modified": "2026-04-29 16:42:58",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
  }
]

```

```

    "applied_transformations": "format_standardization",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "f6aba42225aaa0f82aaf34923eee63bc3560643874003797650cb572f3aae76e"
  },
  {
    "dataset_name": "ingestion_summary.csv",
    "relative_path": "data\\bronze\\retailrocket\\ingestion_summary.csv",
    "stage": "bronze",
    "source_system": "retailrocket",
    "file_format": "csv",
    "size_kb": 0.33,
    "last_modified": "2026-04-29 16:43:31",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "format_standardization",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "a2f54f1e11f15f6793a04b13eb8714a4d17af128702d2c94f39b8cbaeb5d946e"
  },
  {
    "dataset_name": "item_properties.parquet",
    "relative_path": "data\\bronze\\retailrocket\\item_properties.parquet",
    "stage": "bronze",
    "source_system": "retailrocket",
    "file_format": "parquet",
    "size_kb": 388430.34,
    "last_modified": "2026-04-29 16:43:30",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "format_standardization",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "c6f67926b226ab85a69a05038f2fa86d511100a271022c0a30e15f0ee23d2d54"
  },
  {
    "dataset_name": "retailrocket_item_snapshot.csv",
    "relative_path": "data\\prepared\\retailrocket_item_snapshot.csv",
    "stage": "prepared",
    "source_system": "retailrocket",
    "file_format": "csv",
    "size_kb": 144479.37,
    "last_modified": "2026-04-29 16:51:43",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "cleaning_preprocessing",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "6579ca26bc0f4a8c339cce8134b67b2bd8e09b6e43809da553f36b900fc4be83"
  },
  {
    "dataset_name": "categories_prepared.csv",
    "relative_path": "data\\prepared\\categories_prepared.csv",
    "stage": "prepared",
    "source_system": "unknown",
    "file_format": "csv",
    "size_kb": 0.32,
    "last_modified": "2026-04-29 16:51:43",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "cleaning_preprocessing",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "d32b783abb9c08c26b14b907e9723d41d040218e616b74c44c9b96a77360a20c"
  },
  {
    "dataset_name": "interactions_prepared.csv",
    "relative_path": "data\\prepared\\interactions_prepared.csv",
    "stage": "prepared",
    "source_system": "unknown",
    "file_format": "csv",
    "size_kb": 196440.63,
    "last_modified": "2026-04-29 16:51:27",
    "load_date": null,
    "load_hour": null,
    "ingestion_date_inferred": null,
    "applied_transformations": "cleaning_preprocessing",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
  }

```

```

"sha256": "c67ae83fec75455b18d81f4b352e87e1d9c41190ae523aa73903eae968f5e9c3"
},
{
  "dataset_name": "prepared_interactions_20260429_215146.parquet",
  "relative_path": "data\\prepared\\prepared_interactions_20260429_215146.parquet",
  "stage": "prepared",
  "source_system": "unknown",
  "file_format": "parquet",
  "size_kb": 42988.9,
  "last_modified": "2026-04-29 21:51:47",
  "load_date": null,
  "load_hour": null,
  "ingestion_date_inferred": null,
  "applied_transformations": "cleaning_preprocessing",
  "versioning_tool": "custom_registry",
  "version_label": "unversioned",
  "dvc_pointer_file": null,
  "sha256": "cbb5b9e5847e6446820ec2ac8e23614267c1386f0670e7203655a24891db46d8"
},
{
  "dataset_name": "prepared_interactions_20260429_215229.parquet",
  "relative_path": "data\\prepared\\prepared_interactions_20260429_215229.parquet",
  "stage": "prepared",
  "source_system": "unknown",
  "file_format": "parquet",
  "size_kb": 42988.9,
  "last_modified": "2026-04-29 21:52:30",
  "load_date": null,
  "load_hour": null,
  "ingestion_date_inferred": null,
  "applied_transformations": "cleaning_preprocessing",
  "versioning_tool": "custom_registry",
  "version_label": "unversioned",
  "dvc_pointer_file": null,
  "sha256": "cbb5b9e5847e6446820ec2ac8e23614267c1386f0670e7203655a24891db46d8"
},
{
  "dataset_name": "prepared_interactions_20260429_215532.parquet",
  "relative_path": "data\\prepared\\prepared_interactions_20260429_215532.parquet",
  "stage": "prepared",
  "source_system": "unknown",
  "file_format": "parquet",
  "size_kb": 65323.8,
  "last_modified": "2026-04-29 21:55:34",
  "load_date": null,
  "load_hour": null,
  "ingestion_date_inferred": null,
  "applied_transformations": "cleaning_preprocessing",
  "versioning_tool": "custom_registry",
  "version_label": "unversioned",
  "dvc_pointer_file": null,
  "sha256": "828d5f22a94671566378e704bd3463d80ffcc951215d27b2d9db36708d1b2c2f"
},
{
  "dataset_name": "products_prepared.csv",
  "relative_path": "data\\prepared\\products_prepared.csv",
  "stage": "prepared",
  "source_system": "unknown",
  "file_format": "csv",
  "size_kb": 59.06,
  "last_modified": "2026-04-29 16:51:43",
  "load_date": null,
  "load_hour": null,
  "ingestion_date_inferred": null,
  "applied_transformations": "cleaning_preprocessing",
  "versioning_tool": "custom_registry",
  "version_label": "unversioned",
  "dvc_pointer_file": null,
  "sha256": "4a3b4bccf86132af617ed578f29b4042b6b2fdb6c7a21b3447c96610fdf7fa3d"
},
{
  "dataset_name": "categories_raw.json",
  "relative_path": "data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\categories_raw.json",
  "stage": "raw",
  "source_system": "dummyjson",
  "file_format": "json",
  "size_kb": 3.13,
  "last_modified": "2026-04-29 16:42:35",
  "load_date": "2026-04-29",
  "load_hour": "11",
  "ingestion_date_inferred": "2026-04-29",
  "applied_transformations": "initial_ingestion",
  "versioning_tool": "custom_registry",
  "version_label": "unversioned",
  "dvc_pointer_file": null,
  "sha256": "1de838fe8b52b789af9076368df35a7b470a3b2f71f5e317ccac1470d2d6fab0"
},
{

```

```

    "dataset_name": "products_raw.json",
    "relative_path":
"data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\products_raw.json",
    "stage": "raw",
    "source_system": "dummyjson",
    "file_format": "json",
    "size_kb": 416.89,
    "last_modified": "2026-04-29 16:42:35",
    "load_date": "2026-04-29",
    "load_hour": "11",
    "ingestion_date_inferred": "2026-04-29",
    "applied_transformations": "initial_ingestion",
    "versioning_tool": "custom_registry",
    "version_label": "unversioned",
    "dvc_pointer_file": null,
    "sha256": "163cfad0fd544bac8058061ea149cf119b1fce3e4edaf679b72e367cd4333e20"
},
{
    "dataset_name": "category_tree.csv",
    "relative_path": "data\\raw\\retailrocket\\load_date=2026-04-29\\lo

```

## 14. Versioning / Lineage Code or Asset Preview

### Asset: *.dvc*

Path: *.dvc*

```
Could not preview file: [Errno 13] Permission denied: 'C:\\Users\\barath\\recomart-
pipeline\\.dvc'
```

### Asset: *config*

Path: *.dvc\\config*

```
[core]
  remote = localstore
['remote "localstore"']
  url = C:\\dvc-storage
```

### Asset: *.dvcignore*

Path: *.dvcignore*

```
# Add patterns of files dvc should ignore, which could improve
# the performance. Learn more at
# https://dvc.org/doc/user-guide/dvcignore
```

### Asset: *prepared.dvc*

Path: *data\\prepared.dvc*

```
outs:
- md5: 2ef466e3f1b1dd72a1d5cc75e632dc55.dir
  size: 329
  nfiles: 2
  hash: md5
  path: prepared
```

### Asset: *raw.dvc*

Path: *data\\raw.dvc*

```
outs:
- md5: 714ee410c1e3626b5fd8049461795db2.dir
  size: 446229
  nfiles: 4
  hash: md5
  path: raw
```

## 15. Validation / Pipeline Log Preview

```
{
  "cells": [
    {
      "cell_type": "code",
      "execution_count": 2,
      "id": "fef32579-f890-4b53-8d24-2fcae64a4914",
      "metadata": {},
      "outputs": [
        {
          "name": "stdout",
          "output_type": "stream",
          "text": [
            "PROJECT_ROOT: C:\\Users\\barath\\recomart-pipeline\\n",
            "RAW_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\n",
            "BRONZE_ROOT: C:\\Users\\barath\\recomart-pipeline\\data\\bronze\\n",
            "\\n",
            "Discovered files:\\n",
            "- events_csv: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-28\\load_hour=14\\events.csv\\n",
            "- category_tree_csv: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-26\\load_hour=19\\category_tree.csv\\n",
            "- item_properties_part1_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-28\\load_hour=14\\item_properties_part1.csv\\n",
            "- item_properties_part2_csv: C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-28\\load_hour=14\\item_properties_part2.csv\\n",
            "- products_raw_json: C:\\Users\\barath\\recomart-
```

```

pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\products_raw.json\\n",
    "- categories_raw_json: C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-28\\load_hour=14\\categories_raw.json\\n",
    "- products_parquet: C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet\\n",
    "- categories_parquet: C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet\\n",
    "\\n",
    "Initial status: FAIL\\n",
    "Final status: PASS\\n",
    "Initial issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260428T144031Z.csv\\n",
    "Initial summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260428T144031Z.csv\\n",
    "Revalidation issues file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_revalidated_20260428T144031Z.csv\\n",
    "Revalidation summary file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_revalidated_20260428T144031Z.csv\\n",
    "Fix log file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\fix_log_20260428T144031Z.csv\\n",
    "JSON report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.json\\n",
    "PDF report file: C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260428T144031Z.pdf\\n"
    ]
    },
    ],
    "source": [
        "# DM4ML - Assignment - Validation + Auto-Fix + Revalidation + PDF Report\\n",
        "\\n",
        "import json\\n",
        "from pathlib import Path\\n",
        "from datetime import datetime, timezone\\n",
        "\\n",
        "import pandas as pd\\n",
        "\\n",
        "from reportlab.lib import colors\\n",
        "from reportlab.lib.pagesizes import A4\\n",
        "from reportlab.lib.styles import getSampleStyleSheet\\n",
        "from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer, Table, TableStyle,
PageBreak\\n",
        "\\n",
        "# =====\\n",
        "# RecoMart Validation - robust file discovery + validation\\n",
        "# with remediation, revalidation, and PDF reporting\\n",
        "# =====\\n",
        "\\n",
        "PROJECT_ROOT = Path.cwd().resolve()\\n",
        "while not (PROJECT_ROOT / \"src\").exists() and PROJECT_ROOT != PROJECT_ROOT.parent:\\n",
        "    PROJECT_ROOT = PROJECT_ROOT.parent\\n",
        "\\n",
        "RAW_ROOT = PROJECT_ROOT / \"data\" / \"raw\"\\n",
        "BRONZE_ROOT = PROJECT_ROOT / \"data\" / \"bronze\"\\n",
        "\\n",
        "REMIEDIATED_RAW_ROOT = PROJECT_ROOT / \"data\" / \"remediated\" / \"raw\"\\n",
        "REMIEDIATED_BRONZE_ROOT = PROJECT_ROOT / \"data\" / \"remediated\" / \"bronze\"\\n",
        "\\n",
        "REPORT_ROOT = PROJECT_ROOT / \"reports\" / \"validation\"\\n",
        "LOG_ROOT = PROJECT_ROOT / \"logs\"\\n",
        "\\n",
        "for folder in [REPORT_ROOT, LOG_ROOT, REMEDIATED_RAW_ROOT, REMEDIATED_BRONZE_ROOT]:\\n",
        "    folder.mkdir(parents=True, exist_ok=True)\\n",
        "\\n",
        "UTC_NOW = datetime.now(timezone.utc)\\n",
        "RUN_ID = UTC_NOW.strftime(\"%Y%m%dT%H%M%S\")\\n",
        "RUN_TS = UTC_NOW.isoformat()\\n",
        "\\n",
        "LOG_FILE = LOG_ROOT / f\"validation_log_{RUN_ID}.json\"\\n",
        "INITIAL_ISSUES_FILE = REPORT_ROOT / f\"validation_issues_initial_{RUN_ID}.csv\"\\n",
        "INITIAL_SUMMARY_FILE = REPORT_ROOT / f\"validation_summary_initial_{RUN_ID}.csv\"\\n",

```

## 16. Data Quality Report Preview

```

{
    "run_id": "20260429T111342Z",
    "run_ts": "2026-04-29T11:13:42.777701+00:00",
    "project_root": "C:\\Users\\barath\\recomart-pipeline",
    "initial_validation": {
        "run_id": "20260429T111342Z",
        "run_ts": "2026-04-29T11:13:42.777701+00:00",
        "phase": "initial",
        "project_root": "C:\\Users\\barath\\recomart-pipeline",
        "raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\raw",
        "bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\bronze",
        "discovered_files": {
            "events_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\events.csv",

```



```

"category_tree_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\retailrocket\\load_date=2026-04-29\\load_hour=11\\category_tree.csv",
"item_properties_part1_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part1.csv",
"item_properties_part2_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\raw\\retailrock
et\\load_date=2026-04-29\\load_hour=11\\item_properties_part2.csv",
"products_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\products_raw.json",
"categories_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\raw\\dummyjson\\load_date=2026-04-29\\load_hour=11\\categories_raw.json",
"products_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\products.parquet",
"categories_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\bronze\\dummyjson\\categories.parquet"
},
"overall_status": "FAIL",
"issues_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_initial_20260429T111342Z.csv",
"summary_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_initial_20260429T111342Z.csv",
"datasets": [
{
"phase": "initial",
"dataset": "retailrocket_events",
"rows": 2756101,
"columns": 5,
"errors": 0,
"warnings": 2734104,
"status": "PASS"
},
{
"phase": "initial",
"dataset": "retailrocket_category_tree",
"rows": 1669,
"columns": 2,
"errors": 0,
"warnings": 25,
"status": "PASS"
},
{
"phase": "initial",
"dataset": "retailrocket_item_properties",
"rows": 20275902,
"columns": 4,
"errors": 0,
"warnings": 0,
"status": "PASS"
},
{
"phase": "initial",
"dataset": "dummyjson_products_raw",
"rows": 194,
"columns": 27,
"errors": 0,
"warnings": 92,
"status": "PASS"
},
{
"phase": "initial",
"dataset": "dummyjson_categories_raw",
"rows": 24,
"columns": 3,
"errors": 1,
"warnings": 0,
"status": "FAIL"
},
{
"phase": "initial",
"dataset": "products_parquet",
"rows": 194,
"columns": 32,
"errors": 0,
"warnings": 0,
"status": "PASS"
},
{
"phase": "initial",
"dataset": "categories_parquet",
"rows": 24,
"columns": 8,
"errors": 0,
"warnings": 0,
"status": "PASS"
}
]
},
"revalidation": {
"run_id": "20260429T111342Z",
"run_ts": "2026-04-29T11:13:42.777701+00:00",

```

```

"phase": "revalidated",
"remediated_raw_root": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw",
"remediated_bronze_root": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\bronze",
"discovered_files": {
  "events_csv": "C:\\Users\\barath\\recomart-pipeline\\data\\remediated\\raw\\events.csv",
  "category_tree_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\category_tree.csv",
  "item_properties_part1_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\item_properties_part1.csv",
  "item_properties_part2_csv": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\item_properties_part2.csv",
  "products_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\products_raw.json",
  "categories_raw_json": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\raw\\categories_raw.json",
  "products_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\bronze\\products.parquet",
  "categories_parquet": "C:\\Users\\barath\\recomart-
pipeline\\data\\remediated\\bronze\\categories.parquet"
},
"overall_status": "PASS",
"issues_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_issues_revalidated_20260429T111342Z.csv",
"summary_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\validation_summary_revalidated_20260429T111342Z.csv",
"fix_log_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\fix_log_20260429T111342Z.csv",
"datasets": [
  {
    "phase": "revalidated",
    "dataset": "retailrocket_events",
    "rows": 2755641,
    "columns": 5,
    "errors": 0,
    "warnings": 2733184,
    "status": "PASS"
  },
  {
    "phase": "revalidated",
    "dataset": "retailrocket_category_tree",
    "rows": 1669,
    "columns": 2,
    "errors": 0,
    "warnings": 25,
    "status": "PASS"
  },
  {
    "phase": "revalidated",
    "dataset": "retailrocket_item_properties",
    "rows": 20275902,
    "columns": 4,
    "errors": 0,
    "warnings": 0,
    "status": "PASS"
  },
  {
    "phase": "revalidated",
    "dataset": "dummyjson_products_raw",
    "rows": 194,
    "columns": 27,
    "errors": 0,
    "warnings": 92,
    "status": "PASS"
  },
  {
    "phase": "revalidated",
    "dataset": "dummyjson_categories_raw",
    "rows": 0,
    "columns": 1,
    "errors": 0,
    "warnings": 0,
    "status": "PASS"
  },
  {
    "phase": "revalidated",
    "dataset": "products_parquet",
    "rows": 194,
    "columns": 32,
    "errors": 0,
    "warnings": 0,
    "status": "PASS"
  },
  {
    "phase": "revalidated",
    "dataset": "categories_parquet",
    "rows": 24,
    "columns": 8,
    "errors": 0,
    "warnings": 0,

```

```

        "status": "PASS"
    }
  ],
  "pdf_report_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260429T111342Z.pdf"
}

```

## 17. Validation Event Log Preview

```

{"event_ts": "2026-04-29T11:13:42.785703+00:00", "stage": "validation", "status": "started",
"message": "Validation started", "run_id": "20260429T111342Z"}
{"event_ts": "2026-04-29T11:14:14.271184+00:00", "stage": "remediation", "status": "started",
"message": "Validation failed; starting remediation", "run_id": "20260429T111342Z"}
{"event_ts": "2026-04-29T11:17:44.036779+00:00", "stage": "validation", "status": "completed",
"message": "Validation workflow completed", "run_id": "20260429T111342Z", "initial_status":
"FAIL", "final_status": "PASS", "pdf_report_file": "C:\\Users\\barath\\recomart-
pipeline\\reports\\validation\\data_quality_report_20260429T111342Z.pdf"}

```

## 18. Data Versioning Event Log Preview

```

{"event_ts": "2026-04-29T17:03:15.276191+00:00", "stage": "lineage", "status": "started",
"message": "Building data versioning and lineage artifacts", "extra": {}}
{"event_ts": "2026-04-29T17:03:24.861004+00:00", "stage": "lineage", "status": "completed",
"message": "Lineage registry and workflow documentation created", "extra": {"registry_csv":
"data_versioning\\registry\\dataset_lineage_registry.csv", "registry_json":
"data_versioning\\registry\\dataset_lineage_registry.json", "workflow_md":
"data_versioning\\docs\\versioning_workflow.md"}}

```

## 19. Conclusion

This PDF consolidates the Data Versioning and Lineage deliverables by documenting dataset locations, inferred versions, lineage metadata, repository structure, and a reproducible versioning workflow suitable for submission.